

Version 2 - solution

1. [7 marks] Use the rules of inference to show that the following argument is valid. For each step, state the rule used and the step(s) to which it applies (if applicable).

$$a \vee b \vee c$$

$$x \wedge y \wedge \neg z$$

$$b \rightarrow (\neg x \wedge c)$$

$$(c \rightarrow m) \wedge (k \rightarrow b)$$

$$\therefore \neg k$$

direct
proof

proof
by
contr.

STEPS		REASON
1	$x \wedge y \wedge \neg z$	given
2	x	C.S. 1
3	$x \vee \neg c$	D.A. 2
4	$\neg(\neg x \wedge c)$	De Morgan's 3
5	$b \rightarrow (\neg x \wedge c)$	given
6	$\neg b$	M.T. 4, 5
7	$(c \rightarrow m) \wedge (k \rightarrow b)$	given
8	$k \rightarrow b$	C.S. 7
9	$\neg k$	M.T. 6, 8
10	k	negate concl.
11	$(c \rightarrow m) \wedge (k \rightarrow b)$	given
12	$k \rightarrow b$	C.S. 11
13	b	M.P. 10, 12
14	$b \rightarrow (\neg x \wedge c)$	given
15	$\neg x \wedge c$	M.P. 13, 14
16	$\neg x$	C.S. 15

$$17 \quad x \wedge y \wedge \neg z$$

$$18 \quad x$$

$$19 \quad \therefore \neg k$$

$$\text{given}$$

$$\text{C.S. 17}$$

lines 16, 18 are in contrad.

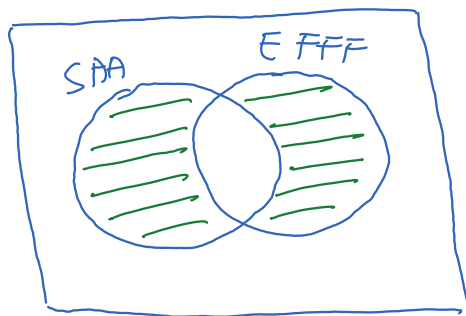
2. [2 marks] Give a set of specific values for a , b , and c that shows that the following argument is **invalid**. Justify your answer.

$$\begin{array}{lcl}
 a \wedge b & 1 \wedge 1 = 1 & \\
 a \vee (b \rightarrow c) & 1 \vee (1 \rightarrow 1) = 1 & \\
 (\neg b \vee c) \rightarrow a & (0 \vee 1) \rightarrow 1 = 1 & \\
 \hline
 \therefore \neg(c \vee b) & \neg 1 \wedge \neg 1 = 0 \wedge 0 = 0 &
 \end{array}$$

$a=1 ; b=1 ; c=1$
 makes all premises true and
 the conclusion false

3. [3 marks] What is the probability that a randomly selected **hex** string of length 12 starts with AA or ends with FFF, but not both?

$$\begin{array}{l}
 \text{AA} \text{ ----- } 16^{10} \\
 (16)(16) \cdot \cdot \cdot (16) \\
 \text{----- FFF } 16^9 \\
 (16)(16) \cdot \cdot \cdot (16) \\
 \text{AA} \text{ ----- FFF } 16^7 \\
 (16)(16) \quad \quad (16)
 \end{array}$$



$$P_r = \frac{(16^{10} - 16^7) + (16^9 - 16^7)}{16^{12}}$$

4. [3 marks] A student has 60 identical Halloween treats.

a) In how many ways can she distribute them to 6 different friends?

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 60$$

$$n=6 \quad r=60 \quad \binom{n+r-1}{r} = \binom{6+60-1}{60} = \binom{65}{60}$$

b) How many distributions in part a) have one friend receiving more than half of the treats?

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 60$$

Making sure that x_1 gets at least 31 treats, give

x_1 31 treats first ; $r = 60 - 31 = 29$

$n=6$
 $r=29$ ← distributing remaining treats

$$\binom{n+r-1}{r} = \binom{6+29-1}{29} = \binom{34}{29}$$

$$\therefore \binom{34}{29} \cdot 6$$

that friend can
be $x_1, x_2, \dots, x_5, x_6$

5. [9 marks] The System Administrator has set the following rules for the password:

- ✓ The password is a string of length 12
- ✓ Only the following 36 characters can be used:

26 letters: A, B, C, D, ..., X, Y, Z
 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9

- ✓ Repetition is allowed unless otherwise stated

- a) [3 marks] How many different passwords, with four digits and eight letters, have no adjacent digits?

Diagram illustrating the arrangement of 8 letters (L) and 4 digits (D) with no adjacent digits. The letters are represented by blue 'L's and the digits by green 'D's. Red arrows indicate the positions where digits can be placed (before, between, and after the letters).

$$26^8 \cdot \binom{9}{4} \cdot 10^4$$

to place the digits

$$\therefore \binom{9}{4} (26)^8 (10)^4$$

- b) [3 marks] How many different passwords have at least five letters and at least five digits?

5L and 7D or 6L and 6 digits or 7L and 5D
 LLLLL DDDDDDD LLLLLL DDDDD LLLLLL DDDDD

$$\frac{12!}{5!7!} (26)^5 (10)^7 + \frac{12!}{6!6!} (26)^6 (10)^6 + \frac{12!}{7!5!} (26)^7 (10)^5$$

- c) [3 marks] How many different passwords, with letters only, have some letter present five times, a different letter present four times, and three other distinct letters?

$L_1 L_1 L_1 L_1 L_1 L_2 L_2 L_2 L_2 L_3 L_4 L_5$

$$\frac{12!}{5!4!} \cdot \binom{26}{1} \binom{25}{1} \binom{24}{3}$$

select L_1 select L_2 select L_3, L_4, L_5

6. [4 marks] Circle one answer only; you MUST justify the answer. No marks will be given for answers without justification. No marks will be given when justification does not imply the correctness of the answer.

Universe $U = \{-6, -5, -3, -2, -1, 0, 1, 2, 3, 5\}$

$A = \{-3, -2, 1, 2, 3\}$

$B = \{-2, -1, 1, 2\}$

$C = \{-6, -5, -2, 0, 1, 2, 6\}$

a) $\exists a \in A, \forall b \in B$ then $a \cdot b \notin C$

TRUE / FALSE

a	b	$a \cdot b \notin C$
-3	-2	$6 \notin C$ x
-2	-1	$2 \notin C$ x
1	1	$1 \notin C$ x
2	1	$2 \notin C$ x
3	2	$6 \notin C$ x

$\{-6, -5, -1, 0, 6\}$ $\{-3, -1, 3\}$

b) $\exists a \in (\overline{B \cap C}), \forall b \in (\overline{A \cap B})$ then $a \cdot b \in C$

TRUE / FALSE

a	b	$a \cdot b \in C$
0	-3	$0 \in C$ ✓
0	-1	$0 \in C$ ✓
0	3	$0 \in C$ ✓

7. [4 marks] Consider the following programming segment. Your answer must rely on combinations structure. Answers that use sigma notation will not be accepted.

Determine the value of the variables `counter1`, and `counter2` after the segment is executed.

Do not simplify the expressions.

(Note: the for loops include both the start and end values)

```
counter1 = 11
counter2 = 22
for i = b+1 to 3*n do {
    counter1 = counter1 + 33
    for j = i+1 to 5*n do {
        counter2 = counter2 + 44
    }
}
//assume b and n are positive integers with n>b
```

Answer:

$$\text{counter1} = 11 + (3n - (b+1) + 1)(33) = 11 + (3n - b)(33)$$

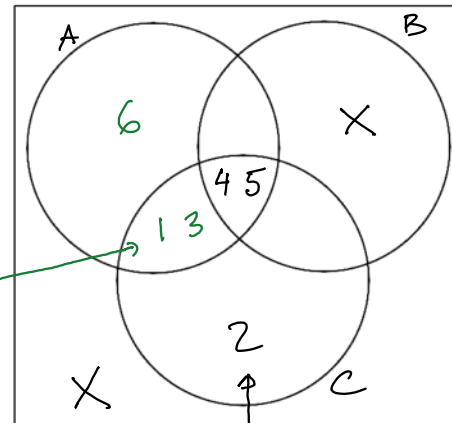
$$\text{counter2} = 22 + \left[\binom{5n - (b+1) + 1}{2} - \binom{5n - 3n}{2} \right] (44)$$

$$= 22 + \left[\binom{5n - b}{2} - \binom{2n}{2} \right] (44)$$

8. [5 marks] Provide one example of nonempty sets A, B and C that satisfies all of the following:

- $(B - C) \subseteq A$
- $C - (A \cup B) = \{2\}$
- $\{4, 5\} \subseteq \overline{(B \cap C) \Delta A}$
- $\overline{(A - B)} \subseteq (B \cup C)$
- $|B \cup (\overline{C} \cap A)| = 3$

Two el
are
here



Universe $U = \{1, 2, 3, 4, 5, 6\}$.

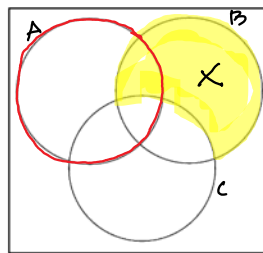
Write your answer:

$$A = \{1, 3, 4, 5, 6\}$$

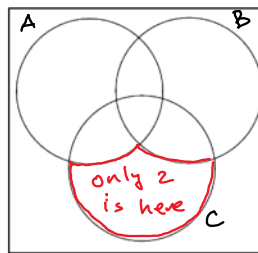
$$B = \{4, 5\}$$

$$C = \{1, 2, 3, 4, 5\}$$

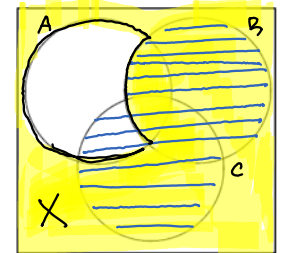
Must write your answer here!
Leaving at least one set empty
implies zero marks.



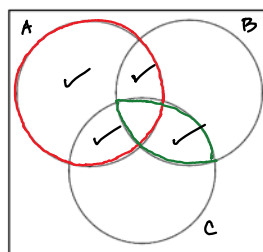
$$(B - C) \subseteq A$$



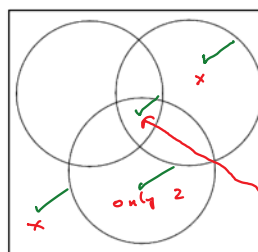
$$C - (A \cup B) = \{2\}$$



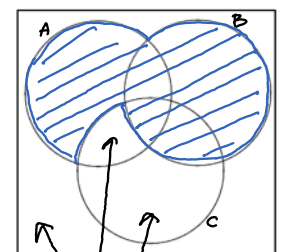
$$(A - B) \subseteq (B \cup C)$$



$$(B \cap C) \Delta A$$



$$\{4, 5\} \subseteq (B \cap C) \Delta A$$



$$B \cup (A \cap C) \quad 3 \text{ el. are here}$$

$$B \cup (\overline{C} \cap A)$$

complement

4, 5 are here